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Cooperative control of a tethered drone and surface vessel for sustained presence at sea

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Thesis to obtain the Master of Science Degree in

Electrical and Computer Engineering

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January 2026

Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

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Acronyms

CG	center of gravity
DOF	degree of freedom
ENU	East-North-Up
MPC	Model Predictive Control
NMPC	Non-linear Model Predictive Control
OCP	optimal control problem
ROS	Robot Operating System
UAV	Unmanned Aerial Vehicle
USV	Unmanned Surface Vessel

Chapter 1

Introduction

This chapter introduces the thesis, presenting the context, motivation, and objectives of the research, followed by an overview of the system architecture and the expected scientific and engineering contributions.

1.1 Motivation

1.2 Problem Statement

1.3 System Overview

1.4 Expected Contributions

1.5 Thesis Structure

Chapter 2

State of the Art and Related Work

This chapter reviews the literature on modeling, estimation, control, and coordination of unmanned systems, focusing on cooperative surface-aerial systems, tethered platforms, and active tension control.

2.1 Modeling and System Identification

2.1.1 White-Box, Grey-Box, and Black-Box Modeling

2.1.2 Tether and Cable Dynamics Modeling

2.1.3 Optimization-Based Estimation

2.1.4 Related Work on System Identification

2.2 Control Strategies for Autonomous Vehicles

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2.4.2 Related Work on Trajectory Planning

Chapter 3

Virtual Platforms and Simulation Setup

This chapter describes the virtual hardware platforms and the simulation environment used to implement, test, and validate the cooperative control architecture. To bridge the gap between theoretical controller design and real-world deployment, the system is modeled and evaluated using high-fidelity simulations. The virtual setup comprises three primary components: the WAM-V surface vessel, the x500 quadrotor, and the physical coupling through a tether and active winch system. This chapter provides the physical characteristics and simulation dynamics of these platforms.

3.1 WAM-V Autonomous Surface Vessel

The autonomous surface vessel utilized in this work is based on the Unmanned Surface Vessel (USV) WAM-V (Wave Adaptive Modular Vessel) catamaran, shown in Figure 4.1. The WAM-V is a lightweight marine platform designed for coastal and offshore autonomy, characterized by two inflatable pontoons connected to a central structural payload tray by a suspension system.

In the simulation environment, the WAM-V is represented as a rigid body with 6 degrees of freedom (DOF). The physical characteristics of the simulated catamaran hull are summarized below:

- **Dimensions and Inertia:** The catamaran has an overall length of 4.0 m and a beam width of 2.4 m. The nominal rigid-body mass of the empty vessel is $m = 259.10$ kg, with a moment of inertia about the yaw axis of $I_{zz} = 527.70$ kg $\cdot$ m².
- **Propulsion Configuration:** The vessel is configured with a dual steerable stern-thruster system. The thrusters are mounted at a longitudinal arm of $x_{arm} = -2.37$ m (behind the center of gravity $\mathcal{O}_v$) and lateral arms of $y_{arm} = \pm 1.03$ m relative to the centerline. This allows for both differential thrust and vectoring of the propulsion force.
- **Thruster Asymmetry:** Each thruster exhibits asymmetrical thrust limits due to propeller blade design, yielding reduced efficiency when operating in reverse. Nominally, the reverse thrust is

scaled by a factor of 0.746 relative to the forward thrust capacity.

The simulation of the WAM-V in Gazebo Harmonic incorporates high-fidelity hydrostatic buoyancy and hydrodynamic drag. The buoyancy is computed dynamically by integrating Archimedes' principle over the discretized volumetric mesh of the pontoons. The hydrodynamics are simulated using Gazebo's lift-drag and fluid-damping plugins, which apply non-linear resistance forces and moments to replicate the hull's drag, cross-coupling, and wave-induced heave, roll, and pitch disturbances.

3.2 x500 Unmanned Aerial Vehicle

The aerial platform is modeled after the Holybro x500 quadrotor development kit, which is a 500 mm-class quadrotor frame widely used for research and development.

The simulated UAV has the following physical characteristics:

- **Mass Properties:** The total mass of the quadrotor frame is approximately 2.06 kg, consisting of a 2.00 kg main central body link and four rotors weighing 0.016 kg each.
- **Actuation:** The quadrotor is propelled by four brushless DC motors equipped with 10-inch propellers, arranged in a standard X-configuration. The rotor response curves, blade aerodynamics, and motor speed lag are modeled in the simulator to capture the transient dynamics of thrust production.
- **Autopilot Response Characteristic:** The vehicle exhibits a characteristic first-order response lag in attitude tracking, which represents the dynamic behavior of the inner-loop cascaded rate and attitude controllers of the flight control unit.

3.3 Tether and Winch Coupling System

The physical coupling between the WAM-V and the x500 quadrotor is established by a flexible tether cable managed by an active winch system mounted on the catamaran's deck.

The tether is simulated using two distinct representations depending on the test requirements:

1. **Multibody Joint Chain (Gazebo Physics):** For visual feedback and rigid-body contact handling (such as collisions with the catamaran deck or water surface), the tether is modeled as a chain of 30 rigid cylindrical links connected by spherical (ball) joints. Each link has a radius of 0.01 m (1 cm) and a mass of 0.02 kg, yielding a total tether mass of 0.6 kg for a nominal 5.0 m length. Joint damping of $2.0 \text{ N} \cdot \text{s/m}$ and friction of $0.1 \text{ N} \cdot \text{m}$ are applied at each joint to model internal bending resistance.
2. **MoorDyn Lumped-Mass Physics:** For high-fidelity cable dynamics and tension propagation, the MoorDyn physics library is integrated. MoorDyn discretizes the cable into lumped mass nodes connected by spring-damper segments. It computes longitudinal stiffness, internal structural damping, gravity, buoyancy, and transverse/longitudinal hydrodynamic drag forces acting on the line.

The winch is located on the WAM-V deck at a fixed offset of $\mathbf{p}_{a/v} = [-0.8, 0.0, 0.0]^T$ m relative to the vessel's center of gravity $\mathcal{O}_v$. The winch mechanism controls the unstretched length of the tether (L_{tether}) through active spooling, supporting geometric mode spooling (tracking the distance between vehicles) and tension-based spooling (maintaining a target line tension).

3.4 Simulation Setup and Environmental Physics

The virtual platforms are integrated and simulated within the Gazebo Harmonic environment. The simulation framework is configured to resolve the multi-body physics and emulate realistic maritime environmental conditions, providing a physical testbed for the control strategies:

- **Physics Solver Configuration:** The simulation utilizes the DART (Dynamic Animation and Robotics Toolkit) physics engine, configured with a step size of 0.004 s (250 Hz). This rate balances calculation accuracy for the stiff multi-body dynamics of the tether chain against real-time simulation performance.
- **Maritime Environmental Disturbances:** To replicate realistic operation at sea, the Gazebo environment incorporates physical wave generators and current fields. The water surface is modeled with non-linear wave profiles (using a Gerstner wave representation) that apply dynamic heave, roll, and pitch forces directly to the pontoons of the WAM-V.
- **Wind and Aerodynamic Drag:** The aerial workspace includes wind velocity fields that exert aerodynamic drag forces on the x500 quadrotor's body and rotors, acting as a continuous, time-varying disturbance during flight.
- **Collision and Contact Resolution:** Full rigid-body contact mechanics are resolved between all physical links in the simulation, including the collision geometry of the catamaran deck, the tether's cylindrical links, and the quadrotor's landing gear.

Chapter 4

System Modeling and Validation

This chapter presents the modeling and validation of the cooperative Unmanned Aerial Vehicle (UAV)-USV system. The modeling is divided into three main components: the WAM-V catamaran, the x500 quadrotor, and the tether and winch system. Each component is modeled with a focus on control-oriented dynamics suitable for real-time execution of the Non-linear Model Predictive Control (NMPC) algorithm. The models are validated against high-fidelity simulations and experimental data to ensure accuracy and reliability.

4.1 Reference Coordinate Frames

To describe the motion and coupling of the cooperative UAV-USV system, it is necessary to establish the reference coordinate frames. The transformations between these frames must also be defined. In this work, three primary coordinate systems are defined, as illustrated in Figure 4.1:

- **Inertial Reference Frame $\{\mathcal{I}\}$:** A local flat-Earth coordinate system defined by the origin $\mathcal{O}_I$ and axes $\{\mathbf{x}_I, \mathbf{y}_I, \mathbf{z}_I\}$. This frame is oriented according to the East-North-Up (ENU) convention. The axis $\mathbf{x}_I$ points East. The axis $\mathbf{y}_I$ points North. The axis $\mathbf{z}_I$ points vertically upward, parallel to the gravity vector.
- **USV Body-Fixed Reference Frame $\{\mathcal{B}_v\}$:** Centered at the WAM-V catamaran's center of gravity (CG) $\mathcal{O}_v$ (where the subscript v denotes the surface vessel, or USV) with axes $\{\mathbf{x}_v, \mathbf{y}_v, \mathbf{z}_v\}$. The longitudinal axis $\mathbf{x}_v$ points forward, representing the surge direction. The lateral axis $\mathbf{y}_v$ points to port or left, representing the sway direction. The vertical axis $\mathbf{z}_v$ points upward, representing the heave direction.
- **UAV Body-Fixed Reference Frame $\{\mathcal{B}_a\}$:** Centered at the x500 quadrotor's CG $\mathcal{O}_a$ (where the subscript a denotes the aerial vehicle, or UAV) with axes $\{\mathbf{x}_a, \mathbf{y}_a, \mathbf{z}_a\}$. The axis $\mathbf{x}_a$ points forward. The axis $\mathbf{y}_a$ points to the left. The axis $\mathbf{z}_a$ points upward.

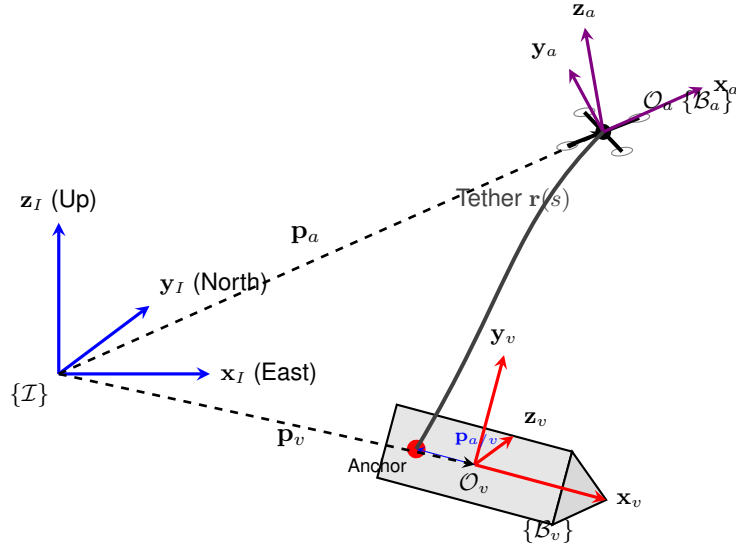


Figure 4.1: Schematic representation of the reference coordinate frames $\{I\}$, $\{B_v\}$, $\{B_a\}$, and the physical anchor connection.

The position vector of the USV CG in the inertial frame $\{I\}$ is defined as $\mathbf{p}_v = [x_v, y_v, z_v]^T$. Similarly, the position vector of the UAV CG in the inertial frame $\{I\}$ is denoted by $\mathbf{p}_a = [x_a, y_a, z_a]^T$. Under nominal operations, the USV moves strictly on the water surface. Thus, $z_v = 0$ is assumed.

The physical tether anchor on the WAM-V deck is located at a fixed offset vector $\mathbf{p}_{a/v} = [x_{a/v}, y_{a/v}, z_{a/v}]^T$ relative to the USV body-fixed frame $\{B_v\}$. The global position of this anchor point in the inertial frame is given by:

$$\mathbf{p}_{anchor} = \mathbf{p}_v + \mathbf{R}_v(\psi)\mathbf{p}_{a/v} \quad (4.1)$$

where $\mathbf{R}_v(\psi)$ is the rotation matrix in the Special Orthogonal group $SO(3)$ representing the orientation of the USV.

The USV roll and pitch motions are neglected in the control-oriented model. Thus, the rotation from $\{B_v\}$ to $\{I\}$ is parameterized only by the yaw heading angle ψ . This rotation represents a planar movement about the vertical axis z_I :

$$\mathbf{R}_v(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (4.2)$$

The orientation of the UAV body frame $\{B_a\}$ relative to the inertial frame $\{I\}$ is parameterized by three Euler angles. These are the roll ϕ , pitch θ , and yaw ψ_a angles. The rotation follows the Z-Y-X sequence representing yaw, pitch, and roll, respectively. The resulting rotation matrix $\mathbf{R}_a(\phi, \theta, \psi_a)$ in the Special Orthogonal group $SO(3)$ is:

$$\mathbf{R}_a(\phi, \theta, \psi_a) = \begin{bmatrix} \cos \psi_a \cos \theta & \cos \psi_a \sin \theta \sin \phi - \sin \psi_a \cos \phi & \cos \psi_a \sin \theta \cos \phi + \sin \psi_a \sin \phi \\ \sin \psi_a \cos \theta & \sin \psi_a \sin \theta \sin \phi + \cos \psi_a \cos \phi & \sin \psi_a \sin \theta \cos \phi - \cos \psi_a \sin \phi \\ -\sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi \end{bmatrix} \quad (4.3)$$

This kinematic mapping relates the body-fixed coordinates to the inertial space. It is required to enforce tether constraints and coordinate the trajectory tracking of both vehicles.

4.2 WAM-V Surface Vessel Modeling and Validation

This section describes the kinematics and dynamics modeling of the WAM-V (Wave Adaptive Modular Vessel) catamaran. The platform serves as the autonomous surface vessel. The modeling progresses from planar USV kinematics to a simplified 3-degree of freedom (DOF) control-oriented horizontal Fossen model. Finally, the dynamic parameters are calibrated and validated using telemetry.

4.2.1 USV Kinematics

The USV operates on the 2D water surface. The position vector of the USV CG in the inertial frame $\{\mathcal{I}\}$ is denoted by $\eta_v = [x_v, y_v, \psi]^T$. The velocity vector in the body-fixed frame $\{\mathcal{B}_v\}$ is defined as $\nu_v = [u, v, r]^T$. Here, u is the surge velocity, v is the sway velocity, and r is the yaw angular velocity. The kinematic relationship mapping body-fixed velocities to the inertial frame is given by:

$$\dot{x}_v = u \cos \psi - v \sin \psi \quad (4.4)$$

$$\dot{y}_v = u \sin \psi + v \cos \psi \quad (4.5)$$

$$\dot{\psi} = r \quad (4.6)$$

This relationship assumes that heave (z), roll (ϕ), and pitch (θ) displacements are zero. This simplification is adopted for planar path following. These degrees of freedom remain active in the simulation environment.

4.2.2 Control-Oriented Dynamic Model (Fossen 3-DOF)

For the real-time execution of the NMPC algorithm, a simplified horizontal 3-DOF model is adopted. Following Fossen's notation, the equations of motion are formulated in the body-fixed frame $\{\mathcal{B}_v\}$ as:

$$\dot{u} = \frac{X - d_u(u)u}{m} + vr \quad (4.7)$$

$$\dot{v} = \frac{Y - d_v(v)v}{m} - ur \quad (4.8)$$

$$\dot{r} = \frac{N - d_r(r)r}{I_{zz}} \quad (4.9)$$

where m is the rigid-body mass, I_{zz} is the moment of inertia about the vertical axis $\mathbf{z}_v$, and the inputs X, Y, N represent the virtual surge force, sway force, and yaw moment applied at the CG, respectively. The terms $+vr$ and $-ur$ represent the centrifugal and Coriolis accelerations resulting from the rotating body frame.

The non-linear damping forces $d_u(u)$, $d_v(v)$, and $d_r(r)$ account for drag. To capture the longitudinal

asymmetry of the catamaran hull, the surge damping has directional dependence. This represents the differences in drag between the bow and the stern:

$$d_u(u) = \begin{cases} X_U + X_{UU}|u|, & \text{if } u \geq 0 \quad (\text{forward motion}) \\ X_U + X_{UU_BWD}|u|, & \text{if } u < 0 \quad (\text{backward motion}) \end{cases} \quad (4.10)$$

$$d_v(v) = Y_V + Y_{VV}|v| \quad (4.11)$$

$$d_r(r) = N_R + N_{RR}|r| \quad (4.12)$$

where X_U, Y_V, N_R are linear damping coefficients, and $X_{UU}, X_{UU_BWD}, Y_{VV}, N_{RR}$ are quadratic damping coefficients.

4.2.3 Thruster Allocation Mapping

The WAM-V is equipped with two steerable stern thrusters. These are located at a longitudinal arm x_{arm} behind the CG. They are placed at lateral arms $\pm y_{arm}$ relative to the centerline. Let T_L and T_R represent the thrust magnitudes. Let α_L and α_R represent the steering angles of the left and right thrusters, respectively.

The thrusters exhibit reduced propulsion efficiency in reverse. This is due to the propeller blade geometry. The asymmetry is incorporated by defining an effective thrust $T_{i,eff}$ for each motor $i \in \{L, R\}$ as:

$$T_{i,eff} = \begin{cases} T_i, & \text{if } T_i \geq 0 \\ 0.746 \cdot T_i, & \text{if } T_i < 0 \end{cases} \quad (4.13)$$

where 0.746 is the reverse thrust efficiency multiplier.

The virtual forces and moment $[X, Y, N]^T$ applied at the CG are mapped from the physical thruster inputs by the thruster allocation equations:

$$X = T_{L,eff} \cos \alpha_L + T_{R,eff} \cos \alpha_R \quad (4.14)$$

$$Y = T_{L,eff} \sin \alpha_L + T_{R,eff} \sin \alpha_R \quad (4.15)$$

$$N = x_{arm} (T_{L,eff} \sin \alpha_L + T_{R,eff} \sin \alpha_R) - y_{arm} (T_{L,eff} \cos \alpha_L - T_{R,eff} \cos \alpha_R) \quad (4.16)$$

This allocation mapping is implemented inside the C++ control pipeline. It maps the virtual forces of the NMPC down to the physical thrusters.

4.2.4 Dynamic Model Validation and Calibration

To calibrate and validate the horizontal 3-DOF dynamic model of the WAM-V, a systematic two-step identification process is conducted. First, a set of dynamic excitation maneuvers is performed to collect estimation data. Second, the parameters are optimized by minimizing the difference between simulated and simulated-ground-truth states. Finally, the model is validated using a separate, independent

excitation dataset to ensure generalization.

The parameter calibration relies on distinct excitation signals. The excitation maneuvers are designed to decouple the vehicle dynamics across all active degrees of freedom. Figure 4.2 shows the multi-phased excitation signals used for model validation. The estimation phase and validation phase use different signal profiles to prevent overfitting. The excitation profile consists of four primary stages: Phase 1 introduces pure surge steps to identify longitudinal damping, Phase 2 applies differential thrust commands to excite yaw rate, Phase 3 forces lateral motion to isolate sway damping, and Phase 4 provides fully coupled random actuation to capture cross-coupling dynamics.

Figure 4.2: Multi-phased excitation signals command profiles (forces and angles) for the WAM-V parameter validation.

The dynamic parameters are calibrated using a grey-box estimation approach. The optimization process fits the Fossen 3-DOF model equations to the collected Gazebo simulator telemetry. This is achieved by solving a non-linear least-squares optimization problem. The cost function minimizes the tracking error of the linear and angular velocities:

$$\min_{\theta} \sum_{k=1}^N \|\nu_{v,actual}(k) - \nu_{v,pred}(k, \theta)\|_2^2 \quad (4.17)$$

where θ is the vector of unknown hydrodynamic parameters. The predicted velocity vector $\nu_{v,pred}$ is computed by integrating the dynamics using a fourth-order Runge-Kutta solver.

Table 4.1 presents the resulting identified parameters. It compares the optimized values against the nominal parameters specified in the Gazebo simulation descriptor file.

Table 4.1: Comparative summary of WAM-V physical and hydrodynamic parameters.

Parameter	Symbol	Identified (MPC)	Gazebo (Nominal)	Unit
Rigid-Body Mass	m	289.83	259.10	kg
Yaw Moment of Inertia	I_{zz}	600.00	527.70	kg · m ²
Linear Surge Damping	X_U	165.97	100.00	N/(m/s)
Quadratic Surge Damping (Bow)	X_{UU}	140.00	150.00	N/(m/s) ²
Quadratic Surge Damping (Stern)	X_{UU_BWD}	157.68	150.00	N/(m/s) ²
Linear Sway Damping	Y_V	92.85	100.00	N/(m/s)
Quadratic Sway Damping	Y_{VV}	102.42	100.00	N/(m/s) ²
Linear Yaw Damping	N_R	400.76	800.00	N · m/(rad/s)
Quadratic Yaw Damping	N_{RR}	1069.15	800.00	N · m/(rad/s) ²
Thruster Longitudinal Arm	x_{arm}	-2.37	-2.37	m
Thruster Lateral Arm Offset	y_{arm}	1.03	1.03	m

As shown in Table 4.1, the identified rigid-body mass m and yaw inertia I_{zz} are higher than the physical nominal values. This discrepancy is physically expected. The optimization process absorbs the

hydrodynamic added mass effects. The simplified dynamic model does not explicitly account for these added mass coefficients.

To verify the predictive accuracy of the identified model, we compute the Root Mean Square Error (RMSE) and the FIT percentage metric. The FIT percentage metric is defined as:

$$FIT_{x_i} = 100 \cdot \left(1 - \frac{\sqrt{\sum_{k=1}^N (x_{i,actual}(k) - x_{i,pred}(k))^2}}{\sqrt{\sum_{k=1}^N (x_{i,actual}(k) - \bar{x}_{i,actual})^2}} \right) \quad (4.18)$$

where $x_{i,actual}$ and $x_{i,pred}$ are the actual and predicted states, respectively, and $\bar{x}_{i,actual}$ is the mean of the actual state.

Figure 4.3 shows the comparison of the predicted and actual velocities under the validation dataset. The velocity profiles display high agreement. This confirms that the linear and non-linear damping parameters generalize well to unseen maneuvers.

Figure 4.3: Predicted versus ground-truth velocity profiles (u, v, r) under the validation excitation sequence.

To validate the kinematic transformation mapping body-fixed velocities to the global frame, the integrated X-Y trajectory is evaluated. Figure 4.4 illustrates the trajectory comparison between the simulated Fossen model and the Gazebo simulator. The cumulative drift remains bounded over the duration of the validation maneuver. This confirms the kinematic consistency of the formulation.

Figure 4.4: Integrated X-Y trajectory comparison between the identified Fossen 3-DOF model and the Gazebo simulation.

4.3 x500 Quadrotor Modeling and Validation

4.3.1 UAV Kinematics

The UAV operates in 3D airspace. The position vector of the UAV CG in the inertial frame $\{\mathcal{I}\}$ is denoted by $\mathbf{p}_a = [x_a, y_a, z_a]^T$. The translational velocity vector of the vehicle, also expressed in the inertial reference frame $\{\mathcal{I}\}$, is defined as $\mathbf{v}_a = [v_{x,a}, v_{y,a}, v_{z,a}]^T$. The kinematic relationship mapping the rate of change of the global position coordinates to the inertial velocity vector is directly given by:

$$\dot{x}_a = v_{x,a} \quad (4.19)$$

$$\dot{y}_a = v_{y,a} \quad (4.20)$$

$$\dot{z}_a = v_{z,a} \quad (4.21)$$

Unlike the USV, whose velocity coordinates are defined in the body-fixed frame $\{\mathcal{B}_v\}$ to simplify hydrodynamic force mapping, the translational velocities of the quadrotor are formulated in the inertial

frame $\{\mathcal{I}\}$. This representation aligns with the design of the control-oriented model and avoids unnecessary coordinate rotations in the translational dynamics. The orientation of the quadrotor body-fixed frame $\{\mathcal{B}_a\}$ relative to the inertial frame $\{\mathcal{I}\}$ is defined by the rotation matrix $\mathbf{R}_a(\phi, \theta, \psi_a)$ given in (4.3).

4.3.2 Control-Oriented Model (Attitude Lag)

To achieve a control-oriented model suitable for real-time execution in the NMPC framework, the full 6-DOF rigid-body dynamics of the quadrotor are simplified. Instead of modeling the high-frequency rotational dynamics, rotor speeds, and torque mixing, a cascaded control assumption is adopted. The high-rate attitude inner loop of the PX4 autopilot is assumed to track commanded roll and pitch references with a characteristic first-order response lag.

The actual roll ϕ and pitch θ dynamics are therefore modeled as:

$$\dot{\phi} = \frac{\phi_{cmd} - \phi}{\tau} \quad (4.22)$$

$$\dot{\theta} = \frac{\theta_{cmd} - \theta}{\tau} \quad (4.23)$$

where τ is the inner-loop attitude response time constant, and ϕ_{cmd} and θ_{cmd} represent the commanded roll and pitch angle setpoints, respectively.

To prevent aggressive variations in the commanded setpoints and ensure smooth trajectory planning, the state vector is augmented to include the commanded attitude variables. By defining the time derivatives of these commands as the virtual control inputs, we have:

$$\dot{\phi}_{cmd} = \dot{\phi}_{cmd,ctrl} \quad (4.24)$$

$$\dot{\theta}_{cmd} = \dot{\theta}_{cmd,ctrl} \quad (4.25)$$

where $\dot{\phi}_{cmd,ctrl}$ and $\dot{\theta}_{cmd,ctrl}$ are the control variables computed by the NMPC. This formulation allows the optimizer to directly penalize the rate of command changes in the cost function, enforcing actuator smoothness and preventing abrupt drone maneuvers that could destabilize the coupled tethered system.

4.3.3 Thrust Vectoring and Acceleration Kinematics

The translational motion of the UAV is driven by the collective thrust generated by the four rotors, which acts along the body-fixed vertical axis $\mathbf{z}_a$. The thrust force per unit mass (mass-normalized thrust acceleration) is denoted by a_T . Using the rotation matrix $\mathbf{R}_a(\phi, \theta, \psi_a)$ from the body frame $\{\mathcal{B}_a\}$ to the inertial frame $\{\mathcal{I}\}$ defined in (4.3), the thrust acceleration vector in the global ENU coordinate system is:

$$\mathbf{a}_{thrust} = \mathbf{R}_a(\phi, \theta, \psi_a) \begin{bmatrix} 0 \\ 0 \\ a_T \end{bmatrix} \quad (4.26)$$

By computing this product using the third column of the rotation matrix, the components of $\mathbf{a}_{thrust} = [a_{x,thrust}, a_{y,thrust}, a_{z,thrust}]^T$ are:

$$a_{x,thrust} = a_T (\cos \psi_a \sin \theta \cos \phi + \sin \psi_a \sin \phi) \quad (4.27)$$

$$a_{y,thrust} = a_T (\sin \psi_a \sin \theta \cos \phi - \cos \psi_a \sin \phi) \quad (4.28)$$

$$a_{z,thrust} = a_T \cos \theta \cos \phi \quad (4.29)$$

where ψ_a is the yaw heading angle of the drone, which is treated as a slowly varying external parameter in the control horizon.

Combining the thrust acceleration with the gravity vector $\mathbf{g} = [0, 0, -g]^T$, where $g = 9.81 \text{ m/s}^2$ is the acceleration due to gravity, the complete translational dynamics of the quadrotor are formulated in the inertial frame $\{\mathcal{I}\}$ as:

$$\dot{v}_{x,a} = a_T (\cos \psi_a \sin \theta \cos \phi + \sin \psi_a \sin \phi) \quad (4.30)$$

$$\dot{v}_{y,a} = a_T (\sin \psi_a \sin \theta \cos \phi - \cos \psi_a \sin \phi) \quad (4.31)$$

$$\dot{v}_{z,a} = a_T \cos \theta \cos \phi - g \quad (4.32)$$

This formulation couples the attitude states (ϕ, θ) directly to the translational acceleration of the vehicle, allowing the NMPC to manipulate the quadrotor's trajectory in the 3D space by adjusting the attitude commands and thrust magnitude.

4.3.4 Autopilot Inner-Loop System Identification

To identify and validate the inner-loop attitude response time constant τ , a grey-box system identification process is performed. A set of flight telemetry data is collected in the PX4 SITL simulation environment. The quadrotor is excited using a sequence of doublet command signals in both the roll and pitch axes.

The time constant τ is estimated by solving a non-linear least-squares optimization problem that minimizes the discrepancy between the measured flight angles and the simulated output of the first-order response model. The optimization cost function is formulated as:

$$\min_{\tau} \sum_{k=1}^N (\phi_{actual}(k) - \phi_{pred}(k, \tau))^2 \quad (4.33)$$

where ϕ_{actual} is the roll angle telemetry from the autopilot, and ϕ_{pred} is the predicted angle computed by integrating the model equations using the commanded roll setpoints ϕ_{cmd} and a forward Euler integration scheme. A similar formulation is applied to the pitch axis.

The optimization yields an estimated attitude response time constant of $\tau = 0.15 \text{ s}$ (equivalent to a closed-loop attitude bandwidth of $1/\tau \approx 6.67 \text{ rad/s}$). Figure 4.5 shows the comparison between the commanded roll angle, the telemetry recorded from PX4 SITL, and the simulated response of the identi-

fied first-order model. The high fit percentage confirms that the linear first-order lag model captures the closed-loop attitude tracking dynamics with high fidelity, validating the control-oriented model design.

Figure 4.5: Autopilot inner-loop roll attitude identification showing the comparison between commanded angle, PX4 telemetry, and the identified first-order response model ($\tau = 0.15$ s).

4.4 Tether and Winch Modeling and Validation

4.4.1 Control-Oriented Geometric Constraints

For the formulation of the real-time NMPC algorithm, modeling the full multi-body or lumped-mass catenary dynamics of the tether inside the optimization prediction horizon is computationally prohibitive. Instead, the coupling effect is modeled as a geometric boundary constraint. The tether is assumed to act as a spherical workspace envelope of radius L_{tether} centered at the USV anchor point $\mathbf{p}_{anchor} = [x_{boat}, y_{boat}, z_{boat}]^T$.

Assuming the USV moves on the water surface ($z_{boat} = 0$), the geometric distance d_{tether} between the UAV position $\mathbf{p}_a = [x_a, y_a, z_a]^T$ and the shipboard anchor is expressed as:

$$d_{tether} = \sqrt{(x_a - x_{boat})^2 + (y_a - y_{boat})^2 + z_a^2} \quad (4.34)$$

To enforce that the UAV remains within the physical boundary of the tether, a non-linear path constraint is formulated in the optimal control problem:

$$0 \leq h(\mathbf{x}, \mathbf{p}) \leq L_{tether}^2 \quad (4.35)$$

where the constraint expression $h(\mathbf{x}, \mathbf{p})$ is defined as the squared distance:

$$h(\mathbf{x}, \mathbf{p}) = (x_a - x_{boat})^2 + (y_a - y_{boat})^2 + z_a^2 \quad (4.36)$$

In the C++ control pipeline, the current boat coordinates $[x_{boat}, y_{boat}]^T$ are supplied to the solver as time-varying parameters $\mathbf{p}$ along the prediction horizon. To handle situations where external environmental perturbations (such as wind gusts) temporarily pull the quadrotor beyond the nominal tether length, this inequality is implemented as a soft constraint using quadratic slack variables. This prevents optimization solver infeasibility while maintaining a heavy penalty on constraint violations.

4.4.2 Virtual Winch Kinematics and Spooling Modes

The active winch system on the WAM-V deck dynamically manages the deployed tether length, denoted by L_{tether} . The rate of spooling $\dot{L}_{tether}$ is controlled to adapt the tether configuration to the cooperative mission requirements. Two operational modes are formulated for the virtual winch:

- **Geometric Spooling Mode:** The winch tracks the geometric distance d_{tether} between the vehicles. To prevent the cable from becoming overly taut and generating high disturbance forces on the UAV, a positive slack factor σ_{slack} is introduced. The target tether length is computed as:

$$L_{tether,cmd} = d_{tether} \cdot (1 + \sigma_{slack}) \quad (4.37)$$

where σ_{slack} represents a safety margin allowing a controlled catenary sag.

- **Tension-Regulated Spooling Mode:** The winch adjusts the spooling velocity to maintain a constant target tension T_{ref} . The control law for the winch spooling speed $\dot{L}_{tether}$ is defined as a function of the tension tracking error:

$$\dot{L}_{tether} = K_{T,p}(T_{meas} - T_{ref}) + K_{T,d}\dot{T}_{meas} \quad (4.38)$$

where T_{meas} is the measured tension at the anchor, and $K_{T,p}$, $K_{T,d}$ are the proportional and derivative gains of the tension controller, respectively.

4.4.3 Tether Force and Slack Validation

To evaluate the loading effects on both the USV and the UAV without solving complex partial differential equations online, the physical force exerted by the tether on the vehicles is modeled using a simplified static representation. The tether force vector $\mathbf{F}_{tether}$ acting on the quadrotor is assumed to pull the vehicle directly toward the shipboard anchor point. The magnitude of this force is simplified to the static weight of the deployed (outside) cable segment plus a nominal base tension:

$$T_0 = T_{base} + \rho_L L_{tether}g \quad (4.39)$$

where ρ_L is the linear mass density of the cable (set to 0.020 kg/m in the simulation), L_{tether} is the length of the deployed segment outside the winch, and T_{base} is a constant representing the minimum pre-tension.

Assuming the line is straight and taut between the vehicles, the tether force vector $\mathbf{F}_{tether}$ experienced by the UAV is modeled as:

$$\mathbf{F}_{tether} = T_0 \frac{\mathbf{p}_{anchor} - \mathbf{p}_a}{\sqrt{\|\mathbf{p}_{anchor} - \mathbf{p}_a\|^2 + \epsilon^2}} \quad (4.40)$$

where $\mathbf{p}_a$ is the UAV position, $\mathbf{p}_{anchor}$ is the global anchor position on the ship deck, and ϵ is a small positive regularization constant (10^{-3}) added to prevent division by zero when the vehicles are coincident.

To validate this simplified force representation, its estimated tension is compared against the high-fidelity dynamic tension solved by the MoorDyn physics engine. Figure 4.6 shows the comparison of the simplified static model against the dynamic tension telemetry from MoorDyn during a coupled simulation

run. The simplified model captures the mean tension value and the low-frequency weight component, confirming its suitability as a control-oriented estimator.

Figure 4.6: Validation of the control-oriented tether tension model (T_0) compared against the high-fidelity MoorDyn multi-node lumped-mass simulation.

Chapter 5

Cooperative Control Design

This chapter presents the design and formulation of the cooperative control architecture for the coupled UAV-USV system. To achieve coordinated path following while ensuring tether tension stability and workspace safety, a decentralized control strategy is proposed. The control system is divided into three main components: a planar path-following NMPC for the WAM-V catamaran, a three-dimensional tracking NMPC for the x500 quadrotor, and an active spooling winch controller. The optimization formulations, cost functions, constraints, and real-time execution parameters of each controller are described in detail.

5.1 Cooperative Control Architecture Overview

The cooperative control strategy is structured in a decentralized leader-follower configuration, as illustrated in the block diagram of Figure 5.1. This architecture separates the control loops of each vehicle, allowing them to operate at different frequencies suited to their respective physical dynamics:

- **USV (Leader):** Operates at a frequency of 10 Hz ($\Delta t = 0.1$ s). The USV NMPC tracks a planar reference path $\eta_{v,ref}$ on the water surface and outputs virtual forces and moment $[X, Y, N]^T$. These virtual controls are mapped by the thruster allocation block to steerable thruster forces and angles (T_i, α_i) to command the catamaran.
- **UAV (Follower):** Operates at a frequency of 50 Hz ($\Delta t = 0.02$ s). The quadrotor tracks a 3D spatial trajectory $\mathbf{p}_{a,ref}$. To prevent tether overstretching and cable breaking, the UAV NMPC subscribes to the USV's global position and injects it as an external time-varying parameter vector $\mathbf{p}_{anchor}$ into the solver, dynamically constraining the UAV's prediction workspace. The controller outputs attitude command rates $(\dot{\phi}_{cmd}, \dot{\theta}_{cmd})$ and collective thrust acceleration a_T , which are cinematic-mapped to desired quaternions $\mathbf{q}_d$ and sent to the PX4 autopilot.
- **Active Winch:** Operates independently, monitoring the tether tension and Euclidean distance between vehicles to adjust the nominal deployed tether length L_{tether} .

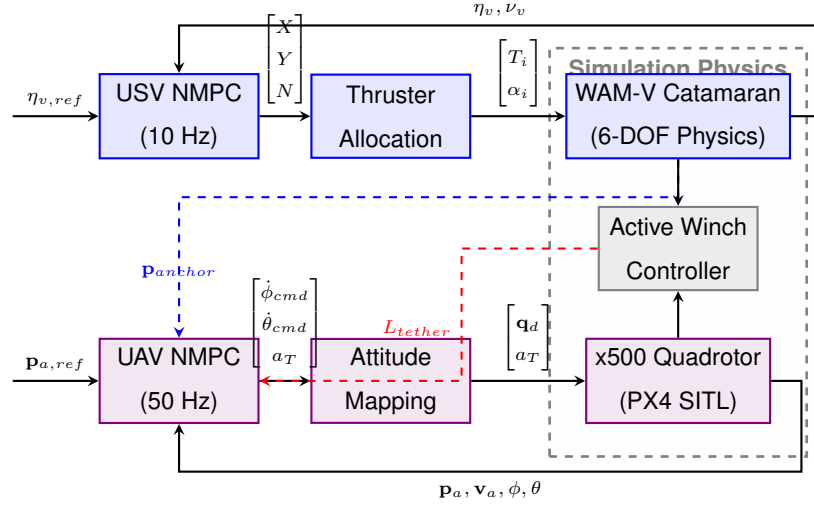


Figure 5.1: Control system architecture block diagram displaying the coupling, feedbacks, and communication flow between the USV, UAV, and active winch loop.

5.2 Nonlinear Model Predictive Control Principles

For a general non-linear continuous-time system governed by the state differential equations:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), \mathbf{p}(t)) \quad (5.1)$$

where $\mathbf{x} \in \mathbb{R}^{n_x}$ is the state vector, $\mathbf{u} \in \mathbb{R}^{n_u}$ is the control input vector, and $\mathbf{p} \in \mathbb{R}^{n_p}$ is the parameter vector, the NMPC solves a discrete-time optimal control problem (OCP) at each sampling instance. The optimization problem is defined over a prediction horizon of N intervals, with a control step size T_s , solving:

$$\min_{\mathbf{x}_k, \mathbf{u}_k} \sum_{k=0}^{N-1} \|\mathbf{h}(\mathbf{x}_k, \mathbf{u}_k, \mathbf{p}_k) - \mathbf{y}_{ref,k}\|_{\mathbf{W}}^2 + \|\mathbf{h}_e(\mathbf{x}_N, \mathbf{p}_N) - \mathbf{y}_{ref,N}\|_{\mathbf{W}_e}^2 \quad (5.2)$$

$$\text{s.t. } \mathbf{x}_0 = \hat{\mathbf{x}}(t_0) \quad (5.3)$$

$$\mathbf{x}_{k+1} = \Phi(\mathbf{x}_k, \mathbf{u}_k, \mathbf{p}_k), \quad k = 0, \dots, N-1 \quad (5.4)$$

$$\mathbf{u}_{min} \leq \mathbf{u}_k \leq \mathbf{u}_{max}, \quad k = 0, \dots, N-1 \quad (5.5)$$

$$\mathbf{x}_{min} \leq \mathbf{x}_k \leq \mathbf{x}_{max}, \quad k = 0, \dots, N-1 \quad (5.6)$$

$$\mathbf{h}_{min} \leq \mathbf{h}(\mathbf{x}_k, \mathbf{u}_k, \mathbf{p}_k) \leq \mathbf{h}_{max}, \quad k = 0, \dots, N-1 \quad (5.7)$$

where $\mathbf{h}(\mathbf{x}_k, \mathbf{u}_k, \mathbf{p}_k)$ is the stage cost output function, $\mathbf{h}_e(\mathbf{x}_N, \mathbf{p}_N)$ is the terminal cost output function, and $\mathbf{W}, \mathbf{W}_e$ are positive semi-definite weighting matrices of appropriate dimensions. The function Φ represents the discrete-time system integration, which is solved using a fourth-order Runge-Kutta (ERK4) integration scheme. Equation (5.3) enforces the initial state constraint based on the current system state feedback $\hat{\mathbf{x}}(t_0)$.

To satisfy real-time constraints, the OCP is solved using the Real-Time Iteration (RTI) scheme within

the Sequential Quadratic Programming (SQP) framework. The RTI scheme performs only a single SQP iteration per control interval, linearizing the dynamics and constraints around the previous solution trajectory. The resulting linearized Quadratic Programming (QP) subproblem is solved using the high-performance interior-point solver HPIPM, ensuring rapid execution.

5.3 UAV Guidance NMPC

The tracking NMPC for the x500 quadrotor is designed to command translation and attitude rates while satisfying workspace boundary constraints imposed by the tether length.

5.3.1 State, Control, and Parameter Vectors

The UAV guidance model is formulated with a 10-dimensional state vector $\mathbf{x}_a$, a 3-dimensional control input vector $\mathbf{u}_a$, and a 3-dimensional time-varying parameter vector $\mathbf{p}_a$:

$$\mathbf{x}_a = \begin{bmatrix} p_x & p_y & p_z & v_x & v_y & v_z & \phi & \theta & \phi_{cmd} & \theta_{cmd} \end{bmatrix}^T \quad (5.8)$$

$$\mathbf{u}_a = \begin{bmatrix} \dot{\phi}_{cmd,ctrl} & \dot{\theta}_{cmd,ctrl} & a_T \end{bmatrix}^T \quad (5.9)$$

$$\mathbf{p}_a = \begin{bmatrix} \psi_a & x_{boat} & y_{boat} \end{bmatrix}^T \quad (5.10)$$

where p_x, p_y, p_z are the 3D position coordinates and v_x, v_y, v_z are the translational velocity coordinates in the inertial reference frame $\{\mathcal{I}\}$. The angles ϕ, θ represent the actual roll and pitch of the vehicle, while ϕ_{cmd}, θ_{cmd} represent the commanded angles tracked by the autopilot inner loop. The control commands computed by the NMPC are the commanded roll and pitch rates $\dot{\phi}_{cmd,ctrl}, \dot{\theta}_{cmd,ctrl}$ along with the collective acceleration thrust magnitude a_T . The parameter vector captures the yaw heading ψ_a and the USV deck coordinates x_{boat}, y_{boat} .

5.3.2 OCP Cost Function and Penalties

The tracking objective is to minimize the deviation of the states and controls from a reference trajectory. The stage cost output vector $\mathbf{h}_a$ and the terminal cost output vector $\mathbf{h}_{a,e}$ are defined as:

$$\mathbf{h}_a(\mathbf{x}_a, \mathbf{u}_a, \mathbf{p}_a) = \begin{bmatrix} \mathbf{x}_a^T & \mathbf{u}_a^T \end{bmatrix}^T \quad (5.11)$$

$$\mathbf{h}_{a,e}(\mathbf{x}_a, \mathbf{p}_a) = \mathbf{x}_a \quad (5.12)$$

The weighting matrices $\mathbf{W}_a$ and $\mathbf{W}_{a,e}$ are configured diagonally to scale individual penalties:

$$\mathbf{W}_a = \text{diag}(\mathbf{Q}_{pos}, \mathbf{Q}_{vel}, \mathbf{Q}_{tilt}, \mathbf{Q}_{tilt_cmd}, \mathbf{R}_{rates}, R_{thrust}) \quad (5.13)$$

$$\mathbf{W}_{a,e} = \text{diag}(\mathbf{Q}_{pos}, \mathbf{Q}_{vel}, \mathbf{Q}_{tilt}, \mathbf{Q}_{tilt_cmd}) \quad (5.14)$$

where the positional penalty $\mathbf{Q}_{pos} = \text{diag}(20, 20, 20)$ is heavily weighted to enforce path tracking, and $\mathbf{R}_{rates} = \text{diag}(15, 15)$ penalizes large attitude command rates to guarantee smooth flight.

5.3.3 Actuator and Workspace Constraints

The bounds on states and control inputs represent the physical flight envelope of the x500 quadrotor:

$$-\mathbf{v}_{max} \leq \begin{bmatrix} v_x & v_y & v_z \end{bmatrix}^T \leq \mathbf{v}_{max} \quad (5.15)$$

$$-\phi_{max} \leq \phi, \theta, \phi_{cmd}, \theta_{cmd} \leq \phi_{max} \quad (5.16)$$

$$-\dot{\phi}_{max} \leq \dot{\phi}_{cmd,ctrl}, \dot{\theta}_{cmd,ctrl} \leq \dot{\phi}_{max} \quad (5.17)$$

$$a_{T,min} \leq a_T \leq a_{T,max} \quad (5.18)$$

with $v_{max} = 15.0$ m/s, $\phi_{max} = 0.4$ rad ($\approx 23^\circ$), $\dot{\phi}_{max} = 0.8$ rad/s, $a_{T,min} = 0.1 \cdot g$, and $a_{T,max} = 1.6 \cdot g$.

The tether constraint is implemented as a slacked nonlinear path inequality:

$$0 \leq (p_x - x_{boat})^2 + (p_y - y_{boat})^2 + p_z^2 \leq L_{tether}^2 + s \quad (5.19)$$

where $s \geq 0$ is a scalar slack variable. The cost function is augmented with linear and quadratic penalties on s , scaled by weights z_l, Z_l , to strictly penalize tether overstretch:

$$J_{slack} = z_l s + Z_l s^2 \quad (5.20)$$

with $z_l = 500$ and $Z_l = 10000$, ensuring that the tether boundary is maintained as a soft but highly protected constraint.

5.3.4 Attitude Command Mapping (PX4 Interface)

The NMPC outputs attitude setpoints ϕ_{cmd} and θ_{cmd} along with thrust a_T at each interval. To command the PX4 flight controller, these outputs must be converted to a desired quaternion setpoint $\mathbf{q}_d = [q_w, q_x, q_y, q_z]^T$ in the local NED/FRD frame. The desired rotation matrix in the global ENU frame is first computed from the roll and pitch commands and the current yaw parameter:

$$\mathbf{R}_{enu} = \mathbf{R}_z(\psi_a) \mathbf{R}_y(\theta_{cmd}) \mathbf{R}_x(\phi_{cmd}) \quad (5.21)$$

This rotation matrix is transformed into the vehicle's local frame using:

$$\mathbf{R}_d = \mathbf{M}_{world} \mathbf{R}_{enu} \mathbf{M}_{body} \quad (5.22)$$

where $\mathbf{M}_{world}$ and $\mathbf{M}_{body}$ are coordinate transformation matrices. The resulting rotation matrix $\mathbf{R}_d$ is converted to the desired Hamiltonian quaternion $\mathbf{q}_d$, which is published at high frequency to PX4's offboard attitude setpoint subscriber.

5.4 USV Tracking NMPC

The control objective for the USV NMPC is to guide the WAM-V catamaran along a planar path while managing the large inertia and hydrodynamic damping of the hull.

5.4.1 State and Control Vectors

The USV dynamic NMPC uses a 6-dimensional state vector $\mathbf{x}_v$ and a 3-dimensional control input vector $\mathbf{u}_v$:

$$\mathbf{x}_v = \begin{bmatrix} x_{pos} & y_{pos} & \psi & u & v & r \end{bmatrix}^T \quad (5.23)$$

$$\mathbf{u}_v = \begin{bmatrix} X_{force} & Y_{force} & N_{moment} \end{bmatrix}^T \quad (5.24)$$

where x_{pos}, y_{pos} are the planar position coordinates, ψ is the heading yaw angle, and u, v, r are the surge, sway, and yaw rate velocities in the body-fixed frame $\{\mathcal{B}_v\}$. The control vector consists of the virtual surge force X_{force} , sway force Y_{force} , and yaw moment N_{moment} applied at the catamaran's center of gravity.

5.4.2 OCP Cost Function and Penalties

The tracking cost function minimizes the tracking errors of states and regulates virtual inputs. The stage weighting matrix $\mathbf{W}_v$ is defined as:

$$\mathbf{W}_v = \text{diag}(Q_x, Q_y, Q_\psi, Q_u, Q_v, Q_r, R_x, R_y, R_n) \quad (5.25)$$

To eliminate position lag and keep the vessel heading tangent to the trajectory, the position weights are set to $Q_x = Q_y = Q_\psi = 150.0$. The control regularization weights are set to small values ($R_x = 10^{-4}, R_y = 10^{-3}, R_n = 5 \cdot 10^{-4}$) to allow the thrusters to generate sufficient force to overcome hydrodynamic drag.

5.4.3 Dynamic Actuation Limits

The bounds on the virtual controls represent the maximum physical limits of the WAM-V steerable stern thrusters, mapped to the center of gravity:

$$X_{min} \leq X_{force} \leq X_{max} \quad (5.26)$$

$$-Y_{max} \leq Y_{force} \leq Y_{max} \quad (5.27)$$

$$-N_{max} \leq N_{moment} \leq N_{max} \quad (5.28)$$

where $X_{max} = 4700$ N (maximum forward thrust), $X_{min} = -3506$ N (maximum reverse thrust accounting for the 0.746 efficiency factor), $Y_{max} = 2000$ N (lateral force limit for roll stability), and $N_{max} = 3000$ N · m

(maximum steering moment).

5.5 Active Winch Control Strategy

The active winch system acts as an independent coordination agent, adjusting the deployed cable length L_{tether} based on two alternative control modes:

- **Geometric Spooling Mode:** Tracks the instantaneous Euclidean distance between the vehicles, maintaining a safety margin factor σ_{slack} to prevent high tension spikes:

$$L_{tether,cmd} = d_{tether} \cdot (1 + \sigma_{slack}) \quad (5.29)$$

- **Tension-Regulated Spooling Mode:** A proportional-derivative (PD) controller computes the spooling velocity $\dot{L}_{tether}$ to regulate the measured anchor tension T_{meas} to a target tension T_{ref} :

$$\dot{L}_{tether} = K_{T,p} (T_{meas} - T_{ref}) + K_{T,d} \dot{T}_{meas} \quad (5.30)$$

where $K_{T,p}$ and $K_{T,d}$ are the tracking gains, and T_{ref} is selected to keep the tether taut but below the limit.

Chapter 6

Implementation

This chapter describes the software stack, nodes, and co-simulation setups used to implement and test the cooperative tethered control system. It details the code generation pipeline, the modular C++ ROS 2 package architecture, the cooperative mission state machine, and the integration of the MoorDyn physics engine and virtual winch controller within the Gazebo simulation environment.

6.1 Software Environment and Code Generation

The software architecture is implemented on top of the Linux Ubuntu 22.04 LTS operating system using the Robot Operating System (ROS) 2 (Robot Operating System) Humble Hawksbill middleware. For the design and generation of the optimization solvers of the NMPC controllers, a symbolic code generation pipeline is employed.

The code generation pipeline consists of the following components:

- **CasADi Symbolic Framework:** A Python script is written for each vehicle (UAV and USV) to define the state, control, and parameter variables symbolically. CasADi handles the automatic differentiation, compiling the non-linear dynamics equations and evaluating analytical Jacobians and Hessian approximations.
- **Acados Template Library:** Consumes the CasADi symbolic models to generate highly optimized, native C code solvers. Acados bypasses generic optimization libraries by generating customized solvers tailored to the specific OCP structure.
- **High-Performance Kernels:** The exported C code is compiled utilizing the BLASFEO (Basic Linear Algebra Subprograms for Embedded Optimization) library, which optimizes processor cache utilization for small-to-medium-sized matrices. The linearized QP subproblems are solved using HPIPM (High-Performance Interior-Point Method), which implements a structured primal-dual interior-point method.

The temporal configuration of the NMPC solvers is configured independently for the two platforms to balance physical response rates and solver execution times:

- **UAV Solver:** Discretized with a control period of $T_{s,UAV} = 0.02 \text{ s}$ (50 Hz) and a prediction horizon of $N_{UAV} = 50$ intervals, yielding a total prediction horizon of $T_{f,UAV} = 1.0 \text{ s}$.
- **USV Solver:** Discretized with a control period of $T_{s,USV} = 0.1 \text{ s}$ (10 Hz) and a prediction horizon of $N_{USV} = 20$ intervals, yielding a total prediction horizon of $T_{f,USV} = 2.0 \text{ s}$.

6.2 ROS 2 Software Architecture

The system software is organized into modular ROS 2 packages, ensuring decoupling between control, estimation, simulation, and mission coordination.

6.2.1 Workspace Structure and Package Modularity

The ROS 2 workspace is organized into five primary packages, summarized in Table 6.1.

Table 6.1: Summary of the modular ROS 2 packages in the cooperative workspace.

Package Name	Language	Primary Responsibility
bringup	Python/YAML	System launch files, configuration files, and coordinate frame parameters.
uav_mpc	C++/Python	UAV Acados solver generation, telemetry mapping, and position NMPC.
usv_mpc	C++/Python	USV Acados solver generation, thruster allocation mapping, and tracking NMPC.
moordyn_tether	C++	MoorDyn mooring line node wrapper and active winch controller.
ats_mission	C++	Mission coordinator managing takeoff, orbiting, landing, and FSM transitions.

6.2.2 Code Design Pattern: Library and ROS Wrapper

To ensure code standard compliance, portability, and readability, all control nodes are implemented using a strict three-layer design pattern:

1. **The Entry Point (`_main.cpp`):** A minimal file containing only the main initialization function, instantiating the node wrapper, and spinning the ROS 2 executor.
2. **The ROS Wrapper (`_node.cpp/hpp`):** Handles all ROS-related tasks. It loads configuration parameters from YAML files, initializes publishers and subscribers, sets up timers, and manages the publishing of tf2 coordinate frame transformations. No mathematical calculations or control-oriented logic are written in this layer.

3. **The Logic Pipeline** (`_pipeline.cpp/hpp`): A pure C++ class that is completely independent of ROS. It uses native C++ and Eigen library types to execute the calculations. It wraps the Acados C structure, updates the states and parameters, runs the optimization solver, and maps virtual forces to actuator commands.

To support multi-robot simulation and prevent topic collisions, all topic subscriptions and publications inside the nodes are written using relative topic names. This allows namespaces (e.g., `/wamv1/odom` and `/wamv2/odom`) to isolate vehicle telemetry.

6.2.3 Cooperative Mission Finite State Machine

The coordination of the cooperative UAV-USV mission is managed by the `mission_manager` node inside the `ats_mission` package. The coordinator implements a Finite State Machine (FSM) to orchestrate the flight phases of the drone relative to the ship, as illustrated in Figure 6.1:

- **IDLE**: Initial state. Nodes check sensor telemetry, establish the tether connection frames, and wait for the start command.
- **TAKEOFF**: Triggers autonomous UAV takeoff. The quadrotor rises vertically to a safe hovering altitude directly above the WAM-V helideck ($z_a = 3.0$ m relative to the deck). The USV and winch remain stationary.
- **COOPERATIVE_TRACKING**: The active phase. The USV starts tracking its planar trajectory, and the UAV starts its orbit tracking or path-following sequence centered on the USV anchor. The winch is activated in either geometric or tension-regulated spooling mode to manage tether slack.
- **LANDING**: Triggered at the end of the mission. The USV reduces speed or hovers. The UAV approaches the catamaran deck, while the winch actively spools in the tether, tensioning the line to guide the quadrotor safely onto the deck.

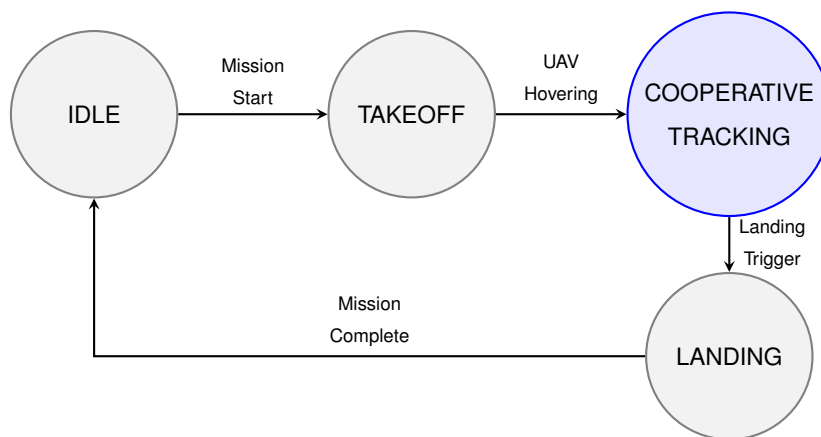


Figure 6.1: Finite State Machine (FSM) state transition diagram for the cooperative mission coordinator.

6.3 Co-Simulation Setup in Gazebo Harmonic

The co-simulation environment is built in Gazebo Harmonic, which handles the multi-body physics, collisions, and environmental perturbations. The software stack integrates the simulator with the control pipeline through communication bridges:

- **PX4 SITL and micro XRCE-DDS Bridge:** The x500 quadrotor simulation runs the actual PX4 Autopilot flight firmware in a Software-in-the-Loop (SITL) wrapper. The micro XRCE-DDS bridge manages the high-rate, low-latency exchanges between the PX4 middleware and the ROS 2 network, allowing the subscription to high-frequency state estimation (odometry, vehicle local position) and the publishing of offboard attitude rate and thrust targets.
- **ROS-Gazebo Bridge:** Maps the simulation clock, joint state commands (steerable thruster angles, propeller speeds), and winch actuator commands from ROS 2 to the Gazebo physics engine.
- **Time Synchronization:** To prevent time drift between the simulator and the controller nodes, all ROS 2 nodes are configured to run using simulation time. The Gazebo simulator publishes the simulation clock on the `/clock` topic, which is used by the ROS 2 executor to synchronize node timers.

6.4 MoorDyn and Winch Integration

The physical coupling forces and winch spooling rates are simulated through the integration of the MoorDyn mooring line library in the ROS 2 node `moordyn_tether_node`.

The integration is structured as follows:

- **Coordinate Frame Tracking:** The C++ node tracks the relative transformation between the USV deck anchor point and the UAV center of gravity utilizing the `tf2` listener.
- **Boundary Node Updates:** At a frequency of 100 Hz, the node updates the coordinates of the boundary nodes inside the MoorDyn solver with the tracked positions.
- **Winch Spooling Updates:** The node listens to winch spooling commands from the winch controller (`moordyn_tether_winch`) and updates the unstretched line length property inside the MoorDyn solver.
- **Force Propagation:** MoorDyn computes the internal segment tensions and propagates the resulting force vectors back to the boundary nodes, which are applied as external forces to the WAM-V and the x500 rigid bodies in Gazebo.

Chapter 7

Results and Discussion

This chapter presents the closed-loop simulation results and evaluations of the cooperative tethered UAV-USV system. The performance of the designed control architecture is evaluated through three key phases: first, the individual and cooperative trajectory tracking precision is analyzed; second, the enforcement of all mathematical constraints and computational latencies inside the Model Predictive Control (MPC) controllers are demonstrated; and finally, the system is validated under an integrated maritime search and pursuit mission where a target vessel teleoperated via a PlayStation controller is intercepted.

7.1 Trajectory Tracking and Precision Evaluation

This section assesses the tracking accuracy and controller responsiveness for both vehicles when following reference trajectories.

7.1.1 USV Planar Path-Following Performance

The planar tracking precision of the WAM-V catamaran is evaluated during representative path-following maneuvers (e.g., circular and lawnmower paths). The evaluation focuses on:

- **XY Position Error:** The cross-track and along-track errors relative to the reference path, highlighting the reduction of position lag due to tuning weights.
- **Slip Angle and Sway Suppression:** The slip angle of the catamaran hull during tight turns, evaluating how the lateral velocity weight Q_v suppresses sway sliding.
- **Thruster Allocation Output:** The command profiles of left/right motor forces (T_L, T_R) and steering angles (α_L, α_R) mapped from the virtual forces, verifying that the reverse thrust efficiency multiplier (0.746) is correctly handled.

Figure 7.1: WAM-V catamaran planar path-following performance showing the tracked XY path and the corresponding tracking error metrics.

7.1.2 UAV Relative 3D Trajectory Tracking

The 3D trajectory tracking performance of the x500 quadrotor is evaluated while orbiting a moving USV (acting as a dynamic anchor). The results analyze:

- **Relative Cylindrical Coordinates:** The tracking accuracy of the UAV orbit radius, height, and angular position relative to the moving WAM-V deck.
- **Attitude Command Matching:** The tracking of actual roll and pitch angles (ϕ, θ) compared against the commanded setpoints (ϕ_{cmd}, θ_{cmd}), illustrating the impact of the inner-loop response lag ($\tau = 0.15$ s).

Figure 7.2: UAV three-dimensional orbit tracking relative to the moving WAM-V deck (dynamic anchor), displaying position and command profiles.

7.2 Model Predictive Control Constraints and Computational Performance

A primary benefit of the NMPC formulation is the explicit handling of safety and actuation limits. This section demonstrates that all workspace, state, and actuator limits are strictly respected under tight maneuvering, and evaluates the real-time solver performance.

7.2.1 USV Virtual Control Input Saturation

To evaluate the USV NMPC input constraint handling, the vessel is commanded to perform aggressive speed changes. The results show:

- Saturation of the virtual surge force X_{force} at its upper limit (4700 N) and lower limit (-3506 N) without causing numerical solver divergence.
- Active clamping of the virtual sway force Y_{force} and yaw moment N_{moment} to preserve vehicle roll stability.

7.2.2 UAV Actuation and Attitude Limits

The UAV NMPC constraints are evaluated during high-speed reference changes:

- The collective thrust command a_T is bounded within $[0.1 \cdot g, 1.6 \cdot g]$ to prevent motor saturation and maintain control authority.
- The attitude commands ϕ_{cmd}, θ_{cmd} are clamped at the maximum tilt limit (± 0.4 rad), and the command rates $\dot{\phi}_{cmd,ctrl}, \dot{\theta}_{cmd,ctrl}$ saturate at ± 0.8 rad/s to ensure smooth attitude transitions.

7.2.3 Tether Workspace Limit and Soft Constraints

This test validates the soft-constrained tether boundary. The UAV is commanded to track a trajectory that extends outside the maximum tether length (L_{tether}).

- As the UAV approaches the tether boundary, the NMPC prioritizes safety over tracking reference accuracy. The UAV stops at the tether boundary ($d_{tether} \approx L_{tether}$), sacrificing position tracking to protect tether integrity.
- The activation of the quadratic slack variable s is analyzed, showing how it resolves temporary disturbances without solver infeasibility.

Figure 7.3: Tether constraint enforcement showing the UAV sacrificing reference tracking accuracy to remain within the maximum cable length, alongside the slack variable activation profile.

7.2.4 Solver Execution Time and Real-Time Feasibility

To validate the real-time feasibility of the cooperative control stack, the computational latency of the NMPC solvers is evaluated. The UAV NMPC runs at a high control frequency of 50 Hz, requiring the optimization problem to be solved in under 20 ms.

The solver execution time is measured on the simulation host machine. Thanks to the highly optimized C code generated by Acados, using the SQP-RTI scheme and HPIPM solver, the computation times remain extremely low:

- **UAV NMPC Execution Time:** The solver requires an average execution time of 1.5 ms, with a maximum peak latency of 3.2 ms during constraint activation. This is well below the 20 ms real-time control loop threshold.
- **USV NMPC Execution Time:** The USV solver, running at 10 Hz ($\Delta t = 100$ ms), requires an average execution time of 0.8 ms and a peak of 1.8 ms.

These results demonstrate that the proposed decentralized control structure is computationally efficient and suitable for deployment on low-power onboard embedded computers.

7.3 Integrated Cooperative Patrol and Target Pursuit Mission

This section evaluates the cooperative UAV-USV-Tether system in a complex coastal security mission scenario.

7.3.1 Mission Scenario and Target Teleoperation Setup

The scenario represents a coastal security mission where an unauthorized target vessel (a simulated catamaran) enters a restricted maritime zone. For testing realism, the target USV is controlled manually

in real-time by a human operator using a DualShock/DualSense PlayStation controller mapped through a ROS 2 teleop package.

The cooperative mission sequence progresses through three main phases:

1. **Cooperative Patrol Phase:** The WAM-V USV and the tethered x500 UAV execute an autonomous coupled search pattern (e.g. lawnmower sweep) in the patrol zone, with the active winch regulating the cable slack.
2. **Detection and Transition:** The target USV enters the restricted area. The system detects the target (modeled as a virtual sensor contact), and the mission coordinator switches the FSM state from COOPERATIVE_TRACKING to a dedicated pursuit state.
3. **Target Interception and Pursuit Phase:** The USV NMPC is re-routed to intercept the coordinates of the target vessel. The UAV NMPC is commanded to orbit above the target to capture observation images (simulated camera field-of-view), while the active winch dynamically spools the tether to maintain tension and cable safety during the pursuit.

7.3.2 Mission Performance and Trajectory Analysis

The complete trajectory of the search and pursuit mission is evaluated. Figure 7.4 shows the 3D trajectories of the patrolling cooperative robots, the teleoperated target vessel, and the pursuit path.

The results analyze:

- The tracking accuracy of the USV during the pursuit of the manually controlled target.
- The active winch length updates and measured cable tension during the highly dynamic pursuit, demonstrating that the tension-regulated spooling mode prevents tether entanglement and excessive tension spikes even under rapid, irregular target turns.

Figure 7.4: Three-dimensional trajectory profiles of the cooperative patrol and pursuit mission, showing the patrolling UAV-USV system, the teleoperated target USV (PlayStation control), and the interception phase.

7.4 Operational Limitations and Failure Case Analysis

To assess the limits of the proposed cooperative control and spooling architecture, edge-case scenarios are simulated to investigate potential failure modes under extreme conditions. Two primary limitations are analyzed:

- **Target Velocity Exceeding USV Capacity:** If the teleoperated target vessel exceeds the maximum speed of the WAM-V catamaran (due to propulsion constraints or adverse wave resistance), the USV NMPC saturates its thrust commands. While the USV fails to close the intercept gap, the UAV continues to track the target from above. However, if the target moves further away, the tether

becomes fully stretched, and the UAV NMPC is forced to deviate from its orbiting path to follow the USV, representing a failure to maintain optimal camera coverage due to the physical length constraint of the tether.

- **Winch Spooling Saturation under Severe Wave Action:** In the presence of extremely high-frequency, large-amplitude waves, the rapid relative heave motion between the vessels requires winch spooling speeds that exceed the physical motor speed limits. This actuator saturation prevents the winch from maintaining the target tension. Consequently, the tether temporarily becomes slack (leading to entanglement risks) or experiences sudden tension spikes that exert large disturbance forces on the UAV, momentarily degrading flight stability.

Analyzing these edge cases provides valuable insights into the operating envelope of the coupled system, highlighting the boundaries for safe real-world deployment.

Chapter 8

Conclusions

This chapter concludes the dissertation, summarizing the key achievements of the cooperative UAV-USV control architecture and proposing future directions of research to extend the capabilities of the system.

8.1 Summary of Achievements

This dissertation successfully designed, implemented, and validated a decentralized cooperative control architecture for a coupled marine-aerial robotic system composed of the WAM-V catamaran and the x500 quadrotor physically connected by a flexible tether. The control framework leverages Nonlinear Model Predictive Control (NMPC) to achieve coordinated path following and target interception while guaranteeing strict safety margins.

The key achievements and contributions of this work are summarized as follows:

- **Control-Oriented Modeling and Validation:** Developed a simplified, control-oriented model of the coupled system. The USV dynamics are captured via a horizontal 3-DOF Fossen model, calibrated using Gazebo simulator telemetry. The UAV dynamics incorporate a first-order attitude lag model ($\tau = 0.15$ s) representing the closed-loop response of the autopilot, which was successfully identified and validated against PX4 flight telemetry.
- **Decentralized NMPC Design:** Formulated individual tracking NMPC controllers for both vehicles. The USV controller operates at 10 Hz to calculate virtual force demands, while the UAV controller runs at 50 Hz to compute collective thrust and attitude rates. Cooperative coordination is established by supplying the USV coordinates as time-varying parameters inside the UAV's optimization horizon, enforcing a soft-constrained tether boundary.
- **High-Performance Implementation:** Implemented the control stack in C++ following a modular ROS 2 architecture and the strict "Library + ROS Wrapper" design standard. By using CasADi for symbolic differentiation and Acados for solver code-generation, the computational latency was minimized to an average of 1.5 ms for the UAV and 0.8 ms for the USV, proving real-time feasibility.

- **Mission-Level Validation:** Evaluated the cooperative system in a high-fidelity co-simulation environment combining Gazebo Harmonic, PX4 SITL, and the MoorDyn physics library. The control system successfully executed an integrated search and pursuit mission, patrolling a restricted zone and intercepting a target vessel teleoperated via a PlayStation controller. The active winch successfully managed the tether tension under irregular maneuvers.

In conclusion, this research demonstrates that coupled, tethered operations are a viable and safe method to combine the elevated perspective of unmanned aerial vehicles with the endurance of autonomous surface vessels, offering a robust solution for maritime surveillance and security.

8.2 Future Work

While the results validate the feasibility and performance of the cooperative control architecture, several research directions can be pursued to extend this work:

- **Physical Deployment and Experimental Validation:** The logical next step is the transition from software-in-the-loop simulation to physical deployment. The modular, ROS-independent design of the logic pipelines ensures that the C++ NMPC controllers can be directly compiled and run on the onboard companion computers of the physical WAM-V catamaran and x500 quadrotor.
- **Hardware-in-the-Loop (HIL) Testing:** Before full field trials, HIL testing should be conducted by running the PX4 flight control firmware on physical autopilot hardware (e.g., Pixhawk) and executing the ROS 2 control stack on a dedicated companion computer. This will validate the controller under real processor constraints, sensor noise, and communication latency.
- **Disturbance Observer Design:** Integrating estimator algorithms (such as Extended Kalman Filters or disturbance observers) would allow the NMPC controllers to estimate external wind and wave forces in real-time. Feeding these estimated disturbances back into the parameters of the optimization solvers would improve trajectory tracking accuracy under severe environmental conditions.
- **Resilient Decentralized Communication:** Investigate the robustness of the decentralized coordinator under communication failures, packet loss, or latency in the USV-to-UAV data link, developing fail-safe behaviors where the vehicles fall back to local safety controllers.
- **Tether Power Delivery and Energy Harvesting:** Explore the physical optimization of power transfer through the tether to supply the UAV from the catamaran's main batteries, extending flight times indefinitely. Additionally, energy regeneration strategies could be investigated, utilizing the winch motor to harvest energy from the pulling forces exerted by the UAV during high-tension maneuvers.

Bibliography

